

Data Mining

- Data mining, *the extraction of hidden predictive information from large databases*,
- The overall goal of the data mining process is to extract information from a data set and transform it into an understandable structure for further use
- Often *Not to be confused with analytics, information extraction, or data analysis*
- But its goal is the extraction of patterns and knowledge from large amount of data, not the extraction of data itself

DATA MINING TECHNIQUES

- It extract interesting patterns such as groups of data records(cluster analysis), unusual records(anomaly detection)
- These patterns can then be seen as a kind of summary of the input data, and may be used in further analysis or, for example, in machine learning and predictive analytics
- For example, the data mining step might identify multiple groups in the data, which can then be used to obtain more accurate prediction results by a decision support system

Data Warehousing

- also known as an **enterprise data warehouse (EDW)**, is a system used for reporting and data analysis. DWs are central repositories of integrated data from one or more disparate sources
- Storing company data in a secondary location which is typically away from production systems.
- Location= where information can be proactively reported and queried against
- Goal =creation of single logical view of data that may reside in many different physical database

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- it exists to help users understand and enhance their organization's performance
- It is designed for query and analysis rather than for transaction processing,
- and usually contains historical data derived from transaction data, but can include data from other sources

TYPES OF DATA WAREHOUSING

- Data Mart
- Online Transaction Processing
- Online Analytical Processing.

Data Mart

- A data mart is a simple form of a data warehouse that is focused on functional area such as sales, finance or marketing
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- Data marts are often built and controlled by a single department within an organization.
- Given their single-subject focus, data marts usually draw data from only a few sources.
- The sources could be internal operational systems, a central data warehouse, or external data.

Online Analytical Processing(OLAP)

- Is characterized by a relatively low volume of transactions. Queries are often very complex and involve aggregations.
- For OLAP systems, response time is an effectiveness measure.
- OLAP databases store aggregated, historical data in multi-dimensional schemas (usually star schemas).
- OLAP systems typically have data latency of a few hours, as opposed to data marts, where latency is expected to be closer to one day.

Online Transaction Processing(OTLP)

- Is characterized by a large number of short on-line transactions (INSERT, UPDATE, DELETE).
- OLTP systems emphasize very fast query processing and maintaining data integrity in multi-access environments.
- For OLTP systems, effectiveness is measured by the number of transactions per second.

- OLTP databases contain detailed and current data.
- The schema used to store transactional databases is the entity model (usually 3NF).

Data mining VS Data warehousing

Data warehouse	Data mining
Process of storing data in order in given dataset	Process of finding pattern in given dataset.
Data warehousing is the process of extracting and storing data to allow easier reporting.	Data mining is the use of pattern recognition logic to identify trends within a sample data set and extrapolate this information against the larger data pool
The tools in data warehousing are designed to extract data and store it in a method designed to provide enhanced system performance	A typical use of data mining is to create targeted marketing programs, identify financial fraud,
Helps in identifying the certain data in a collection of data	Helps in figuring out a certain pattern of a data or a cluster of data

Benefits of Data Warehouse

- **A Data Warehouse Delivers Enhanced Business Intelligence**

By providing data from various sources, managers and executives will no longer need to make business decisions based on limited data

- **A Data Warehouse Saves Time**

Since business users can quickly access critical data from a number of sources—all in one place—they can rapidly make informed decisions on key initiatives

Benefits of Data Warehouse

- **A Data Warehouse Enhances Data Quality and Consistency**
 - Individual business units and departments including sales, marketing, finance, and operations, will start to utilize the same data repository as the source system for their individual queries and reports.
 - Thus each of these individual business units and departments will produce results that are consistent with the other business units within the organization.

Benefits of Data Warehousing

- **A Data Warehouse Provides Historical Intelligence**
 - A data warehouse stores large amounts of historical data so we can analyze different time periods and trends in order to make future predictions
 - can enable advanced business intelligence including time-period analysis, trend analysis, and trend prediction.

Benefits of Data Warehousing

- **A Data Warehouse Generates a High ROI**
 - Return on Investment(ROI)
 - Past references shows that companies that have implemented data warehouses have generated more revenue and saved more money than companies that haven't invested on data warehouses.

Challenges faced on Data Warehousing

- **User expectation**

- end-user demands and expects more accurate and refined results in return of processing,
- however the performance decreases with exploding data and so the efficiency of the system reduces.

- **Systems optimization**

- business intelligence tools require frequent maintenance and fine tuning of whole system in order to meet users' expectations.

Challenges faced on Data Warehousing

- **Data structuring**

- Proper processing of data requires structuring it in a desired format so that further operations can be executed
- As the volume of data increases the task of structuring the unstructured data add-on, slowing down the processing capabilities of system and eventually becomes hectic for the system manager to qualify the data for analytic purpose.

Challenges faced on Data Warehousing

- **Prefabricated vs. Custom warehouse**

- The varieties of warehouses available in market create ambiguity about which type to choose or go for.
- Custom warehouse saves the time of building the warehouses from various operational
- Prefabricated warehouses saves the time of initial configuration and installation.

Challenges faced on Data Warehousing

- **Resource Balancing**

- Many departments inside an organization tend to access the processing capabilities of the warehouse which eventually reduces the performance of the system and decreases the efficiency as the stress on the system increases.
- Access control and security are some techniques which can be used to maintain a balance between the utilization and performance of warehouse systems.